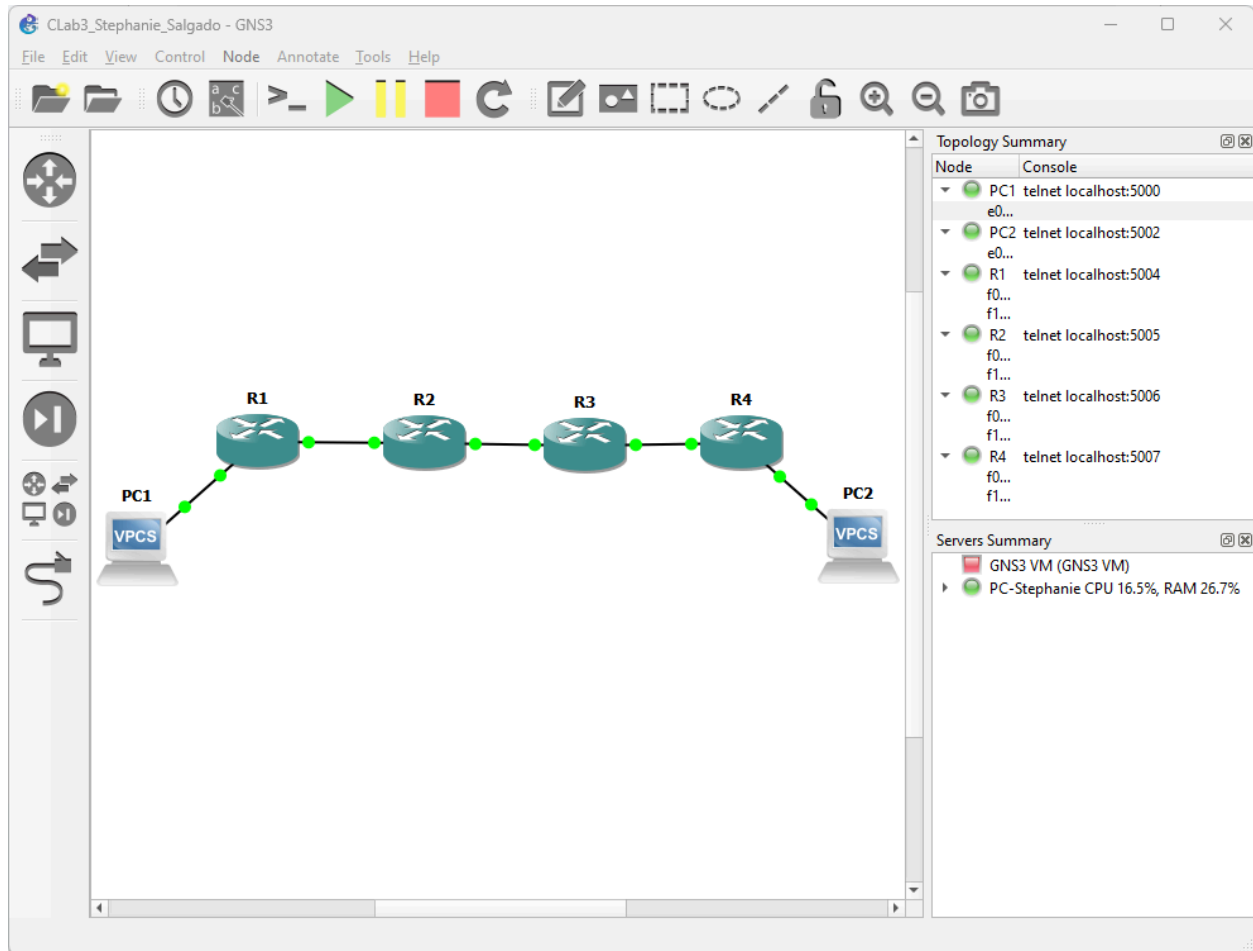


Clab 3

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Creating the network



I created a network with 4 routers and 2 VPCs by dragging them into the workspace and adding a connection between them.

Console connecting to all nodes



PC1

```
PC1> ip 10.10.10.10/24 10.10.10.1
Checking for duplicate address...
PC1 : 10.10.10.10 255.255.255.0 gateway 10.10.10.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> █
```

PC2

```
PC2> ip 50.50.50.50/24 50.50.50.1
Checking for duplicate address...
PC1 : 50.50.50.50 255.255.255.0 gateway 50.50.50.1

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> █
```

Configuring routers:

R1

```
R1#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#ip addr 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#router ospf 1
R1(config-router)#network 10.10.10.0 0.0.0.255 area 0
R1(config-router)#network 20.20.20.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#█
```

To begin I do a basic configuration on R1. After that, I configured OSPF. I do the same for the rest of the routers.

R2

```
R2#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface f0/0
R2(config-if)#ip addr 20.20.20.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface f1/0
R2(config-if)#ip addr 30.30.30.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#router ospf 1
R2(config-router)#network 20.20.20.0 0.0.0.255 area 0
R2(config-router)#network 30.30.30.0 0.0.0.255 area 0
R2(config-router)#exit
```

R3

```
R3#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface f0/0
R3(config-if)#ip addr 30.30.30.2 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface f1/0
R3(config-if)#ip addr 40.40.40.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#router ospf 1
R3(config-router)#network 30.30.30.0 0.0.0.255 area 0
R3(config-router)#
*Sep 17 10:09:54.119: %OSPF-5-ADJCHG: Process 1, Nbr 30.30.30.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config-router)#network 40.40.40.0 0.0.0.255 area 0
R3(config-router)#exit
```

R4

```
R4#configure
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#interface f0/0
R4(config-if)#ip addr 40.40.40.2 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface f1/0
R4(config-if)#ip addr 50.50.50.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#router ospf 1
R4(config-router)#network 40.40.40.0 0.0.0.255 area 0
R4(config-router)#
*Sep 17 10:15:03.079: %OSPF-5-ADJCHG: Process 1, Nbr 40.40.40.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
R4(config-router)#network 50.50.50.0 0.0.0.255 area 0
R4(config-router)#exit
```

Using “show ip route” and “show ip ospf neighbor”:

R1

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.0/24 is directly connected, FastEthernet0/0
L    10.10.10.1/32 is directly connected, FastEthernet0/0
 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.20.20.0/24 is directly connected, FastEthernet1/0
L    20.20.20.1/32 is directly connected, FastEthernet1/0
 30.0.0.0/24 is subnetted, 1 subnets
O    30.30.30.0 [110/2] via 20.20.20.2, 00:05:58, FastEthernet1/0
 40.0.0.0/24 is subnetted, 1 subnets
O    40.40.40.0 [110/3] via 20.20.20.2, 00:05:58, FastEthernet1/0
 50.0.0.0/24 is subnetted, 1 subnets
O    50.50.50.0 [110/4] via 20.20.20.2, 00:01:28, FastEthernet1/0
R1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
30.30.30.1      1    FULL/DR         00:00:35    20.20.20.2    FastEthernet1/0
```

R2

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

 10.0.0.0/24 is subnetted, 1 subnets
O    10.10.10.0 [110/2] via 20.20.20.1, 00:06:08, FastEthernet0/0
 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.20.20.0/24 is directly connected, FastEthernet0/0
L    20.20.20.2/32 is directly connected, FastEthernet0/0
 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.30.30.0/24 is directly connected, FastEthernet1/0
L    30.30.30.1/32 is directly connected, FastEthernet1/0
 40.0.0.0/24 is subnetted, 1 subnets
O    40.40.40.0 [110/2] via 30.30.30.2, 00:08:38, FastEthernet1/0
 50.0.0.0/24 is subnetted, 1 subnets
O    50.50.50.0 [110/3] via 30.30.30.2, 00:01:42, FastEthernet1/0
R2#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
40.40.40.1      1    FULL/BDR        00:00:33    30.30.30.2    FastEthernet1/0
10.10.10.1      1    FULL/BDR        00:00:37    20.20.20.1    FastEthernet0/0
```

R3

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
O       10.10.10.0 [110/3] via 30.30.30.1, 00:05:07, FastEthernet0/0
    20.0.0.0/24 is subnetted, 1 subnets
O       20.20.20.0 [110/2] via 30.30.30.1, 00:07:55, FastEthernet0/0
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.30.30.0/24 is directly connected, FastEthernet0/0
L       30.30.30.2/32 is directly connected, FastEthernet0/0
C       40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.40.40.0/24 is directly connected, FastEthernet1/0
L       40.40.40.1/32 is directly connected, FastEthernet1/0
O       50.0.0.0/24 is subnetted, 1 subnets
O       50.50.50.0 [110/2] via 40.40.40.2, 00:00:37, FastEthernet1/0
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
50.50.50.1       1    FULL/BDR        00:00:32    40.40.40.2     FastEthernet1/0
30.30.30.1       1    FULL/DR         00:00:31    30.30.30.1     FastEthernet0/0
```

R4

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
O       10.10.10.0 [110/4] via 40.40.40.1, 00:00:24, FastEthernet0/0
    20.0.0.0/24 is subnetted, 1 subnets
O       20.20.20.0 [110/3] via 40.40.40.1, 00:00:24, FastEthernet0/0
    30.0.0.0/24 is subnetted, 1 subnets
O       30.30.30.0 [110/2] via 40.40.40.1, 00:00:24, FastEthernet0/0
    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.40.40.0/24 is directly connected, FastEthernet0/0
L       40.40.40.2/32 is directly connected, FastEthernet0/0
C       50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       50.50.50.0/24 is directly connected, FastEthernet1/0
L       50.50.50.1/32 is directly connected, FastEthernet1/0
R4#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
40.40.40.1       1    FULL/DR         00:00:39    40.40.40.1     FastEthernet0/0
```

Notice all the “C” (connected), “L” (local), and “O” (OSPF) codes. In particular R2 and R3, which have 2 neighbors, which makes sense since they are in the middle and connected to 2 different routers.

Testing connectivity:

PC1

```
PC1> ping 50.50.50.50
84 bytes from 50.50.50.50 icmp_seq=1 ttl=60 time=120.756 ms
84 bytes from 50.50.50.50 icmp_seq=2 ttl=60 time=120.256 ms

PC1> ping 10.10.10.1
84 bytes from 10.10.10.1 icmp_seq=1 ttl=255 time=15.375 ms
84 bytes from 10.10.10.1 icmp_seq=2 ttl=255 time=15.377 ms

PC1> ping 20.20.20.2
84 bytes from 20.20.20.2 icmp_seq=1 ttl=254 time=45.225 ms
84 bytes from 20.20.20.2 icmp_seq=2 ttl=254 time=45.793 ms

PC1> ping 30.30.30.2
84 bytes from 30.30.30.2 icmp_seq=1 ttl=253 time=75.140 ms
84 bytes from 30.30.30.2 icmp_seq=2 ttl=253 time=75.179 ms

PC1> ping 40.40.40.2
84 bytes from 40.40.40.2 icmp_seq=1 ttl=252 time=105.245 ms
84 bytes from 40.40.40.2 icmp_seq=2 ttl=252 time=105.744 ms
```

Notice PC1 can successfully ping PC2, and all the routers.

PC2

```
PC2> ping 10.10.10.10
84 bytes from 10.10.10.10 icmp_seq=1 ttl=60 time=121.275 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=60 time=120.719 ms

PC2> ping 10.10.10.1
84 bytes from 10.10.10.1 icmp_seq=1 ttl=252 time=105.246 ms
84 bytes from 10.10.10.1 icmp_seq=2 ttl=252 time=105.533 ms

PC2> ping 20.20.20.2
84 bytes from 20.20.20.2 icmp_seq=1 ttl=253 time=75.286 ms
84 bytes from 20.20.20.2 icmp_seq=2 ttl=253 time=75.232 ms

PC2> ping 30.30.30.2
84 bytes from 30.30.30.2 icmp_seq=1 ttl=254 time=45.158 ms
84 bytes from 30.30.30.2 icmp_seq=2 ttl=254 time=45.167 ms

PC2> ping 40.40.40.2
84 bytes from 40.40.40.2 icmp_seq=1 ttl=255 time=15.226 ms
84 bytes from 40.40.40.2 icmp_seq=2 ttl=255 time=15.182 ms
```

PC2, is also able to ping PC1, and all the routers.

R1 ping R4

```
R1#ping 40.40.40.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 40.40.40.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/90/92 ms
```

R2 ping PC2

```
R2#ping 50.50.50.50
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.50.50.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/75/76 ms
```

R3 ping PC1

```
R3#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/75/76 ms
```

R4 ping R2

```
R4#ping 20.20.20.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.20.20.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 48/63/76 ms
```

Summary

In this lab, I simulated a network with 4 routers and 2 VPCs. I used a basic configuration for all routers and then configured OSPF on them. This time, I noticed the network was much larger than the one we used for RIP. At first I thought the PCs would pose a problem, but they didn't. Instead I noticed that OSPF allowed faster connections between all routers and VPCs. After this lab, I can see why Open Shortest Path First (OSPF) is the more efficient protocol.